

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR
CHEMICAL ENGINEERING DEPARTMENT
END SEMESTER EXAMINATION, Autumn Semester, 2002-2003
Subject : TRANSPORT PHENOMENA
Sub. No. : 144005 Time : 3 hrs Full Marks : 50 No. of Students : 38
OPEN BOOK EXAM., ANSWER ALL QUESTIONS

- 1.(a) For flow over a flat plate with an extremely rough surface, convection heat transfer effects are known to be correlated by the expression

$$Nu_x = 0.04 Re_x^{0.9} Pr^{1/3}$$

For airflow at 50 m/s, what is the surface shear stress at $x = 1$ m from the leading edge of the plate? Assume the air to be at a temperature of 300 K (For air at 300K $\rho = 1.1614 \text{ Kg/m}^3$, kinematic viscosity $= 1.589 \times 10^{-5} \text{ m}^2/\text{s}$, $k = 2.63 \times 10^{-2} \text{ W/m.K}$ $Pr = 0.7$) 5

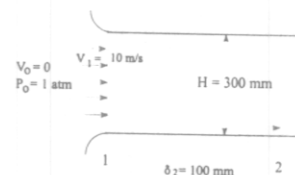
- 1.(b) A thin flat plate is installed in a water tunnel as a splitter. The plate is 0.3 m long and 1 m wide. The freestream speed is 2 m/s. Laminar boundary layer forms on both sides of the plate. The boundary layer velocity profile is approximated by

$$\frac{v_x}{U} = 2 \left[\frac{y}{\delta} \right] - \left[\frac{y}{\delta} \right]^2$$

Obtain an expression for δ . Determine the total viscous drag force on the plate assuming the pressure drag is negligible. (kinematic viscosity of water is $1.0 \times 10^{-6} \text{ m}^2/\text{s}$. 5

2. The record-write head for a computer disk storage system floats above the spinning disk on a very thin film of air (the film thickness is 0.5 micron). The head location is 150mm from the disk centreline, the disk spins at 3600rpm. The record-write head is 10mm square. Determine for standard air (density= 1.23 kg/m^3 , viscosity= $1.78 \times 10^{-5} \text{ kg/(m.sec)}$) in the gap between the head and the disk (a) the Reynold's number of the flow, (b) the viscous shear stress and (c) the power required to overcome viscous shear stress. For such a small gap the flow can be considered as flow between parallel plates. 10

3. Flow of air ($\rho = 1.23 \text{ kg/m}^3$) develops in a flat horizontal duct following a well rounded entrance section. The duct height is $H = 300 \text{ mm}$. Turbulent boundary layer grows on the duct walls, but the flow is not yet fully developed. Assume that the velocity profile in each boundary layer is given by $u/U = (y/\delta)^{1/7}$. The inlet flow is uniform with $V = 10 \text{ m/s}$ at section 1. At section 2, the boundary layer thickness on each wall of the channel is $\delta_2 = 100 \text{ mm}$. Show that for this flow $\delta^* = \delta/8$. Evaluate the static gauge pressure at section 2. Find the average wall shear stress between the entrance and section 2 at $L = 5 \text{ m}$. The region outside the entrance (location O in figure) can be taken to be still air with pressure equal to atmospheric pressure. 10



4. Benzene, a harmful chemical, has been spilled on the laboratory floor and has spread to a length of 2m. If a film 1 mm depth is formed, how long will it take for the Benzene to completely evaporate? Ventilation in the laboratory provides for airflow parallel to the surface at 1 m/s, and the benzene and air are both at 25 C. The mass densities of Benzene in the saturated vapor and liquid states are known to be 0.417 and 900 kg/m³, respectively. Given: For Air: γ (kinematic viscosity) = 15.7×10^{-6} m²/s; For Benzene-Air: $D_{AB} = 0.88 \times 10^{-5}$ m²/s, $Sc = 1.78$. 10
5. (a) Can you ever use modified Reynold's analogy between heat transfer from a flat plate and mass transfer from a curved surface? Explain your answer. 2
5. (b) Show that, if a flat plate, sides a, b in length, is towed through a fluid so that the boundary layer is entirely laminar, the ratio of towing speeds so that the drag force remains constant regardless of whether a or b is in the flow direction is given by

$$U_a / U_b = \sqrt[3]{a / b}$$

where U_a is the freestream velocity if side a is in the flow direction and U_b is the corresponding fluid velocity if b is in the flow direction. 8

Important relations:

<i>Laminar</i> : $\delta = 5 \times Re_x^{-1/2}$ <i>Laminar</i> : $C_{f,x} = 0.664 Re_x^{-1/2}$ <i>Laminar</i> : $Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$ <i>Laminar</i> : $\overline{C_{f,x}} = 1.328 Re_x^{-1/2}$	<i>Turb</i> : $\delta = 0.37 \times Re_x^{-1/5}$ <i>Turb.</i> : $C_{f,x} = 0.0592 Re_x^{-1/5}$ <i>Turb</i> : $Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3}$ <i>Laminar</i> : $\overline{Nu_x} = 0.664 Re_x^{1/2} Pr^{1/3}$
<i>Mixed B. L</i>	
$\overline{C_{f,L}} = 0.074 / Re_L^{1/5} - 1742 / Re_L$	$\overline{Nu_L} = (0.037 Re_L^{4/5} - 871) Pr^{1/3}$

L is the total length and x_c is the length at which transition to turbulent flow takes place. Sherwood number relations can be directly obtained by replacing Pr by Sc.